# Case Investigation Brief: Fertiliser Token

404_Brain_not_found | Priyanshi Nagariya, Aryan Swarnkar, Kamlesh, Mayank Sejwar | AI Synergy Hackathon 2026 | KhadSetu: a token should be a promise, not a queue number.

- Track: [your track, e.g. AI in Agriculture / AI for All]
- Case: a farmer must get a digital token before buying fertiliser; today the token is a queue position, not a guarantee.
- This brief states what we know, what we assume, and what we still must verify.

## 1. Selected theme / case

Fertiliser Token: a farmer must obtain a digital token before buying fertiliser. The token is a queue position and not a guarantee of stock, date or distance. We build KhadSetu, which issues a token only after stock is reserved at a specific centre, on a date before the sowing deadline, at the nearest feasible centre.

## 2. Stakeholders

| Stakeholder | Wants | Pain today | Influence |
|---|---|---|---|
| Landowner farmer | Fertiliser before sowing | Travel, waiting, stock-outs, wrong dates | Low / high collectively |
| Tenant / sharecropper | Same access | No Farmer ID: no token at all | Very low |
| Centre operator | Orderly counters, correct stock | Crowds, disputes, no forecast | Medium |
| Village verifier | Fair, defensible decisions | Pressure, no clear process | Medium |
| District officer | Working system, no unrest | No live demand-vs-stock view | High |
| State fertiliser dept | Policy that holds | Rollout suspended in ~3 months | Highest |
| Suppliers / logistics | Predictable demand | Unclear where stock should go | Medium (upstream) |

## 3. Problem identified

**Symptoms: 30-50 km travel, hours of waiting, tokens dated after the sowing need, stock gone on arrival, repeat trips, some farmers shut out.**

Root causes: (1) tokens issue without checking stock; (2) dates come from queue position, not sowing deadlines; (3) no matching by distance and stock; (4) Farmer-ID gate excludes tenants and one OTP failure blocks access; (5) a 3-district pilot went statewide without scale testing. Separately: a total supply shortage is NOT solved by tokens.

## 4. Alternative problem statements considered

Scored 1-5 (5 best). D totals highest on feasibility, yet we choose A: D never fixes the root cause (unbacked tokens), and A produces the demand signal B needs later.

| Opt | Description | Imp | Feas | AI | Data | Risk | Tot |
|---|---|---|---|---|---|---|---|
| A* | Stock-backed smart token (chosen) | 4 | 4 | 3 | 3 | 3 | 17 |
| B | Upstream stock logistics | 5 | 2 | 4 | 2 | 2 | 15 |
| C | Tenant identity layer only | 3 | 4 | 2 | 3 | 3 | 15 |

| Opt | Description | Imp | Feas | AI | Data | Risk | Tot |
|---|---|---|---|---|---|---|---|
| D | Info app (stock display + SMS) | 3 | 5 | 3 | 3 | 4 | 18 |
| E | Village/doorstep delivery redesign | 5 | 1 | 1 | 2 | 1 | 10 |

## 5. Evidence and assumptions

Status key: BRIEF = from the case brief; ASM = our stated assumption; VERIFY = needs a real source before it is presented as fact.

| Claim | Source | Confidence | Status |
|---|---|---|---|
| 30-50 km travel, hours of waiting | Case brief | Medium (anecdotal) | BRIEF |
| Token ignores stock/date/distance | Case brief | High | BRIEF |
| Tenants lack Farmer ID | Case brief | High | BRIEF |
| One OTP failure can block | Case brief | Medium | BRIEF |
| 3-district pilot, statewide, suspended ~3 months | Case brief | High (fact); cause unknown | BRIEF |
| Small/marginal holding share (India) | Agri Census | - | VERIFY |
| Tenant cultivator share | NSS / Census | - | VERIFY |
| Fertiliser subsidy size | Union Budget | - | VERIFY |
| Rush-week demand clustering | Assumption | Medium | ASM |
| OTP fail 15%, no-show 10% | Assumption | Low | ASM: replaced by pilot data |

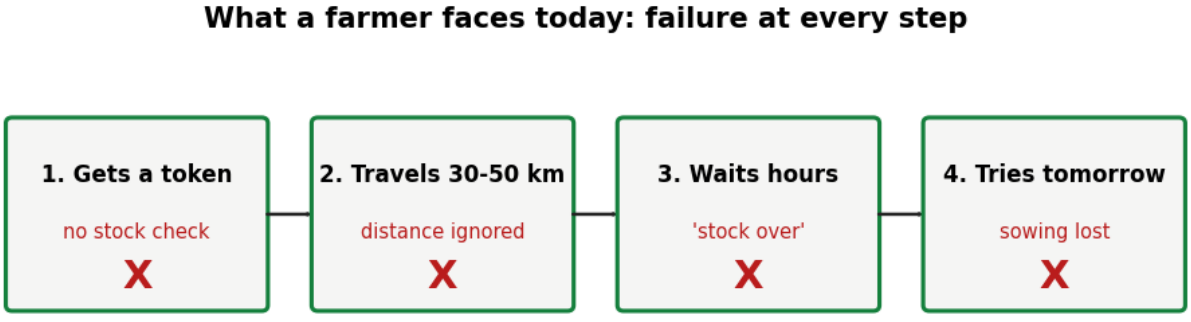
Unknown: real per-centre stock; true suspension cause; exclusion counts. We state the suspension as fact, never its cause.

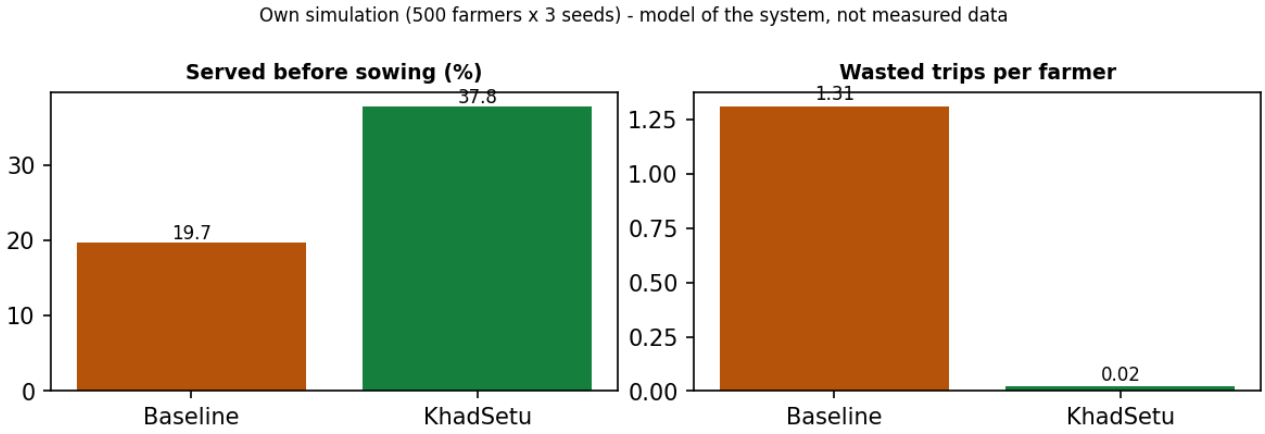
## 6. Why we chose this problem

It fixes the root cause behind every symptom; it is buildable and testable in 36 hours; the simulator gives evidence instead of claims; AI has a real but bounded role; and it creates the demand data option B needs later. D was simpler and scored higher on feasibility, but information without reservation leaves the core failure intact.

## 7. Proposed solution (with measured results)

KhadSetu reserves stock first, matches the sowing date and nearest feasible centre, waitlists honestly with ETAs, includes tenants/OTP failures via audited fallbacks, and tests rules in a simulator. AI helps state needs, forecasts, flags anomalies and explains; rules decide stock, humans decide eligibility. It cannot fix total shortage.






Own simulation (500 farmers x 3 seeds): served-in-window baseline 19.7% vs KhadSetu 37.8%; wasted trips 1.31 vs 0.02. Where KhadSetu does not help is listed in the findings. Intake parser 30/30; forecast MAE 3.92 vs naive 4.31; 100 threads on 50 bags: 0 over-reserved.

Limits, said out loud: no fertiliser is created; baseline is our model, not measured data; OTP/Farmer ID/SMS/stock feeds are mocked; forecaster and anomaly watch train on synthetic data until real data exists.